CCNA 200-301 V2.0

32 AI in Network Operations

Part V — Wireless, Virtualisation and Operations

EXAM TOPICS

5.1 5.2

LECTURER

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THE COURSE

Where this chapter sits

Part I	Foundations	chapters 1–7
Part II	Switching and Network Access	chapters 8–14
Part III	IP Routing	chapters 15–19
Part IV	Network Services and Security	chapters 20–26
Part V	**Wireless, Virtualisation and Operations**	chapters 27–32
Part VI	Sitting the Exam	chapters 33–34

LEARNING OUTCOMES

What you will be able to do

- **Distinguish** generative AI from agentic AI, and describe what makes a system agentic.
- **Describe** where AI is genuinely used in network operations, and where it should not be trusted.
- **Select** an effective prompt, using the four components the blueprint names: data classification, output format, persona and instructions.
- **Judge** what data may safely be sent to an AI system, and sanitise what may not.

ASSUMED KNOWLEDGE

Before we start

Chapter 30 for the five management approaches — agentic systems sit on top of them, not instead of them. Chapter 31 for automation, which is what an agent actually uses to act. Chapter 23 for why credentials must not leave the organisation.

KEY TERMS

Vocabulary for this chapter

- generative AI
- agentic AI
- autonomy
- tool use
- human in the loop
- prompt
- persona
- data classification
- sanitisation
- hallucination
- blast radius

Each is defined where it first matters — not here.

ON THE EXAM

Two topics, and the second one is unusual

This is what the blueprint actually asks for

5.1 is *describe the role of agentic AI in network operations*.

5.2 is *select a prompt to send to a generative AI system to support network operations considering prompt components such as data classification, output format, persona, and instructions*.

Read **5.2** carefully, because it tells you the question format. *Select* means you will be shown several candidate prompts and asked which is best — not asked to write one. And the four components are **named**, which means the marking scheme is those four. A prompt that has all four beats one that does not.

Note the order Cisco lists them in: **data classification comes first**. That is not accidental, and Section sec:dataclass explains why.

Generative and agentic

DEFINITION

Generative AI

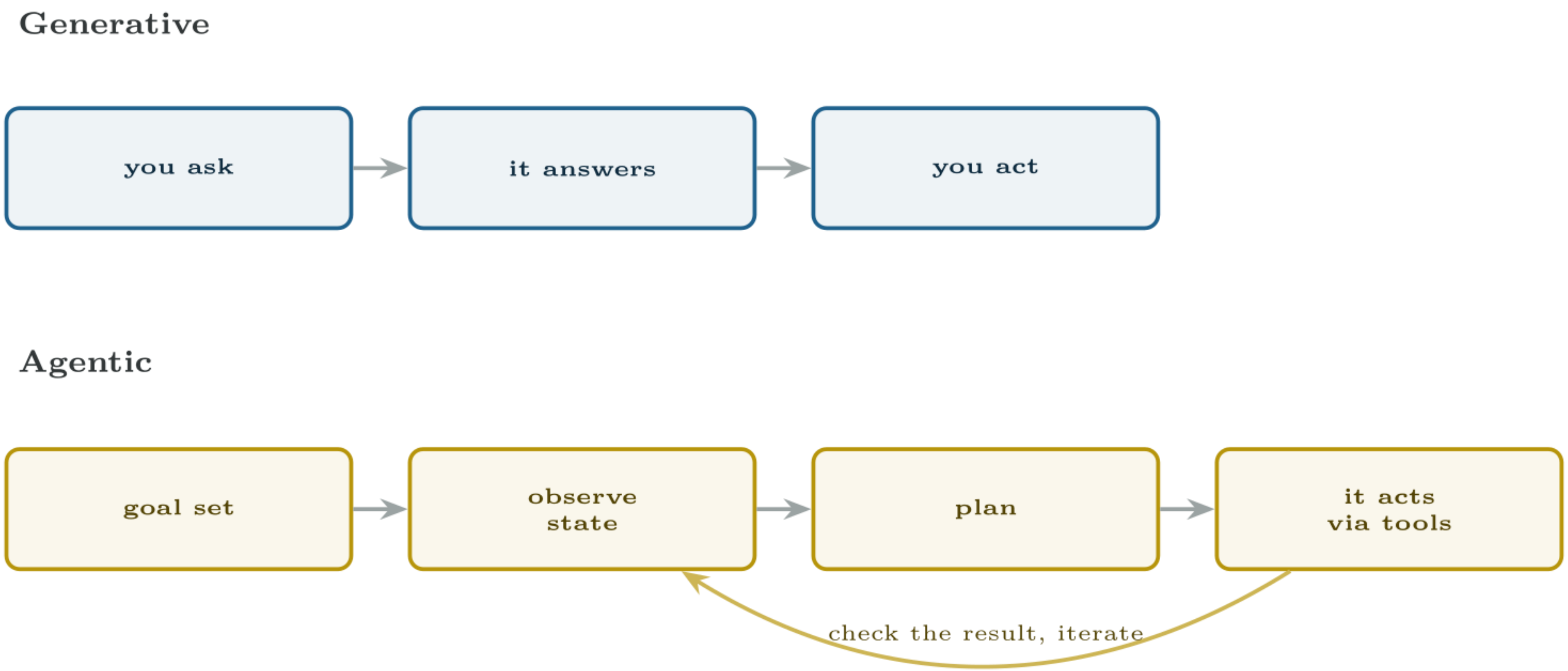
A system that produces content — text, code, configuration — in response to a prompt. It answers and stops. Every action taken as a result is taken by you.

DEFINITION

Agentic AI

A system that pursues a *goal* rather than answering a question. It perceives the current state, plans, uses tools to act, observes the result, and iterates — choosing the intermediate steps itself.

Generative and agentic



The difference is the loop and who closes it. A generative system hands you an answer; an agentic one acts and checks its own work.

WATCH OUT

The one-sentence answer for the exam

Generative AI produces content; agentic AI takes action toward a goal, using tools, and evaluates its own results. Autonomy and tool use are the two words that separate them.

Where it is actually used

CONCEPT

Blast radius is the design question

Before granting autonomy, ask what the worst outcome is if the system is confidently wrong. Clearing one err-disabled port has a blast radius of one port. Changing a route map on a core router has a blast radius of the organisation.

The same technology can sit at different rungs for different actions, and a well-designed deployment does exactly that. “Human in the loop” is not a blanket policy; it is a decision made per action, based on reversibility and reach.

WATCH OUT

What these systems are bad at, and you must know

- **Being confidently wrong.** A fluent, plausible, incorrect answer is the characteristic failure. It does not signal uncertainty the way a colleague would.
- **Your network specifically.** Training data is the internet, not your topology. It does not know that VLAN 40 is the guest network here.
- **Currency.** It may recommend commands deprecated years ago, or miss ones added last year.
- **Accountability.** When a change breaks a site, a person is answerable. Change control, approval and rollback do not become optional because a machine proposed the change.

Selecting a prompt

WORKED EXAMPLE

The same question, asked twice

Weak:

Why is my DHCP not working? Here is my config: [entire running configuration pasted, including `snmp-server community`, `username ... secret` and the site's public addressing]

No persona, no output format, no scope — and it has just sent credentials and the organisation's real addressing to a third party. Even a good answer would not justify that.

Strong:

You are a senior network engineer working on Cisco IOS XE 17.12.

Clients in VLAN 30 receive APIPA addresses. VLANs 10 and 20 on the same distribution switch lease correctly. The DHCP server is central and reached across a routed core. Relevant configuration, with addresses substituted:

```
interface Vlan30 / ip address 10.30.0.1 255.255.255.0
```

List the three most likely causes, most likely first. For each give the single show command that would confirm or eliminate it. Reply as a table with columns Cause, Confirming command, Expected output if this is the fault. Do not propose configuration changes.

All four components are present, and each earns its place: the persona sets the platform, the classification decided that addresses would be substituted and credentials omitted, the instructions bound the scope, and the format makes the answer directly actionable.

How to answer a “select the best prompt” question

This is what the blueprint actually asks for

Score each candidate against the four components and pick the one with most. Then apply three tie-breakers:

- **Eliminate anything containing credentials, keys or real public addressing** immediately, however good the rest of it is. Data classification is listed first for a reason and a prompt that leaks is wrong regardless of its other qualities.
- **Prefer specific over polite.** “Please help me with my network” scores nothing. Length is not the measure — *specificity* is.
- **Prefer a stated output format.** It is the component most often missing from the distractors, so it frequently decides the answer.

WATCH OUT

Assume anything you send leaves your control

A prompt sent to an external service has been disclosed to a third party. It may be logged, retained, reviewed by humans, or used for training, depending on the contract — and deleting it later does not undo the disclosure.

Never send: passwords, hashes, SNMP community strings, pre-shared keys, certificates, API tokens; personal data; real public IP addressing; anything your organisation classifies as confidential or above.

Sanitise first: substitute addresses consistently (documentation ranges `192.0.2.0/24`, `198.51.100.0/24`, `203.0.113.0/24` exist for exactly this), replace hostnames, strip credential lines entirely, and include only the configuration relevant to the question.

And check the policy. Many organisations permit an approved internal or contracted tool and prohibit public ones. Knowing which applies to you is part of the job now.

Common pitfalls

- **Pasting a whole running configuration.** Slower, worse answers, and a disclosure.
- **Confusing generative with agentic.** Tool use and autonomy are the difference.
- **Omitting the output format.** You get an essay where you needed three commands.
- **No persona or platform.** You get generic advice, or IOS syntax for a platform you are not running.
- **Accepting a confident answer without verifying.** Check the command exists on your platform before typing it.
- **Granting autonomy without considering blast radius.**

Common pitfalls (continued)

- **Assuming it knows your network.** It knows the internet.
- **Treating an AI-proposed change as exempt from change control.**

The suggested command was accepted and took the site off the air.

An engineer asks an assistant why a trunk is not carrying VLAN 50 and is told to run `switchport trunk allowed vlan 50` on the uplink. They apply it. Every VLAN except 50 immediately stops working across that link.

The prompt that was used: “VLAN 50 isn’t going over my trunk, what command fixes it?”

The suggested command was accepted and took the site off the air.

What would you check first — and what do you expect to see?

Why it is doing that

Diagnosis — of the answer. The command is real, correctly spelled and valid. It also *replaces* the allowed list rather than adding to it, so a trunk carrying 10, 20 and 30 now carries only 50. The correct command was `switchport trunk allowed vlan add 50` (Chapter 10).

Diagnosis — of the prompt, which is the examinable part. Measure it against the four components:

Prompt engineering as an examinable skill

Platform note. This lab needs no particular tool. Use whatever assistant your institution permits — and **check that policy first**, which is task 1 and is not a formality.

1. Find and read your institution's or employer's policy on sending data to AI services. Write down, in one sentence, what you are permitted to send.
2. Take a real configuration from one of your lab devices. Produce a sanitised version fit to include in a prompt: substitute addressing into documentation ranges, rename hosts, remove every credential. List everything you removed and why.
3. Write the weakest useful prompt you can for a real problem from Chapter 14. Record the answer.
4. Rewrite it with all four components. Record the answer. Compare the two for specificity, correctness and usability.

Prompt engineering as an examinable skill (continued)

5. **Verify both answers on the device.** Note anything suggested that was wrong, deprecated, or not supported on your platform. This step is the one that matters.
6. Ask a question whose correct answer depends on context you deliberately withhold — as in this chapter's Break & Fix. Confirm you can produce a confidently wrong answer, and identify which missing component caused it.
7. Write four candidate prompts for one task, one of which is clearly best, and swap with a colleague. Score each other's against the four components. This is the exam question format.

Prompt engineering as an examinable skill (continued)

8. **Autonomy.** For each of these, place it on the four-rung ladder and justify: clearing an err-disabled port; adding a VLAN to an access switch; changing an OSPF cost on a core link; opening a ticket.
9. **Extension.** Write the one-page guidance your team should follow for using AI assistants on network work — what may be sent, what must be verified, and which actions may ever be automatic.

State the difference between generative and agentic AI in one sentence.

Discuss · then advance

Checkpoint 1 of 6

State the difference between generative and agentic AI in one sentence.

> Generative AI produces content in response to a prompt and stops; agentic AI pursues a goal, uses tools to act, and evaluates its own results.

CHECKPOINT 2 of 6

Name the four prompt components the blueprint lists, and say which is listed first and why.

Discuss · then advance

ANSWER

Checkpoint 2 of 6

Name the four prompt components the blueprint lists, and say which is listed first and why.

Data classification, output format, persona and instructions. **Data classification** is listed first, because a prompt that discloses credentials or real addressing is wrong regardless of how good the rest of it is — and the disclosure cannot be undone.

CHECKPOINT 3 of 6

Give three categories of data that must never be sent to an external AI service.

Discuss · then advance

Checkpoint 3 of 6

Give three categories of data that must never be sent to an external AI service.

Credentials of any kind (passwords, hashes, community strings, pre-shared keys, tokens); personal data; and real public IP addressing or anything the organisation classifies as confidential.

CHECKPOINT 4 of 6

What is blast radius, and how does it govern how much autonomy an agent should have?

Discuss · then advance

ANSWER

Checkpoint 4 of 6

What is blast radius, and how does it govern how much autonomy an agent should have?

> How much breaks if the action is wrong. It governs autonomy because the question is not how capable the system is but how bad a confident mistake would be — clearing one err-disabled port is reversible and local, while changing a core route map reaches the whole organisation.

An assistant gives a fluent, confident, wrong answer. What is this called and what is the defence?

Discuss · then advance

ANSWER

Checkpoint 5 of 6

An assistant gives a fluent, confident, wrong answer. What is this called and what is the defence?

> A hallucination. The defence is verification: check the command exists on your platform, check it against the exhibit or the device, and keep change control. Fluency is not evidence.

CHECKPOINT 6 of 6

Name three genuine uses of AI in network operations.

Discuss · then advance

ANSWER

Checkpoint 6 of 6

Name three genuine uses of AI in network operations.

> Any three of: anomaly detection, event correlation and noise reduction, root-cause suggestion, configuration drafting and review, documentation and summarisation, capacity forecasting, guided remediation.

What to take away

- Generative AI produces content in response to a prompt. **Agentic AI pursues a goal**, uses tools, acts, and evaluates its own results. Autonomy and tool use are the difference.
- Real roles: anomaly detection, event correlation, root-cause suggestion, configuration drafting, documentation, capacity forecasting, guided remediation.
- Autonomy is a ladder — recommend, approve then act, act then report, fully autonomous — chosen per action according to blast radius.
- The four prompt components: **data classification, output format, persona, instructions**. A prompt with all four beats one without.

What to take away (continued)

- Data classification is listed first. Never send credentials, keys, personal data or real public addressing; sanitise into documentation ranges first; check your organisation's policy.
- In a "select the prompt" question: eliminate anything that leaks, prefer specificity over politeness, and look for the stated output format.
- These systems are confidently wrong, do not know your network, and may be out of date. Verify before you apply, and keep change control.

Where we got to

- Generative AI produces content in response to a prompt. **Agentic AI pursues a goal**, uses tools, acts, and evaluates its own results. Autonomy and tool use are the difference.
- Real roles: anomaly detection, event correlation, root-cause suggestion, configuration drafting, documentation, capacity forecasting, guided remediation.
- Autonomy is a ladder — recommend, approve then act, act then report, fully autonomous — chosen per action according to blast radius.
- The four prompt components: **data classification, output format, persona, instructions**. A prompt with all four beats one without.

NEXT

Chapter 33 — A Troubleshooting Method for the 28 Per Cent